

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Templistik®, Part Number 3165163, or equivalent

Additional Service Items

- Torch
- Brass drift
- Hammer

Disassemble

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

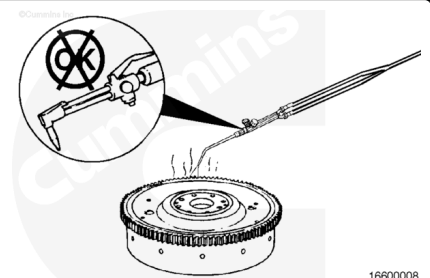
⚠ WARNING ⚠

To reduce the possibility of personal injury or equipment damage, this procedure must only be performed by suitably qualified service technicians.

⚠ WARNING ⚠

To reduce the possibility of personal injury, wear goggles and protective clothing.

⚠ CAUTION ⚠



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Adjust the torch to a good heating flame. Do not use a cutting flame (a blue flame) to heat the ring gear. Damage to the flywheel can result.

⚠ CAUTION ⚠

Do not use a steel drift.

Heat the ring gear with a torch

Use a brass drift and a hammer to remove the hot ring gear.

⚠ WARNING ⚠

To reduce the possibility of personal injury or equipment damage, this procedure must only be performed by suitably qualified service technicians.

⚠ WARNING ⚠

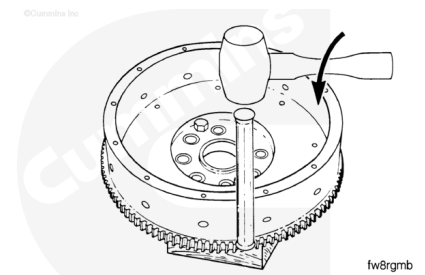
To reduce the possibility of personal injury, wear goggles and protective clothing.

⚠ CAUTION ⚠

Do not use a steel drift.

If a heating torch is **not** available, the ring gear can be removed with a brass drift and hammer.

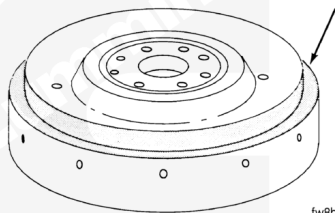
If this method is used to remove the ring gear from a flexplate assembly, make sure a wooden block is used to support the ring gear adapter.



Clean and Inspect for Reuse

Check the flywheel outside diameter for damage at the ring gear location.

If the flywheel is damaged, it **must** be replaced.



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Assemble

⚠ WARNING ⚠

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⚠ WARNING ⚠

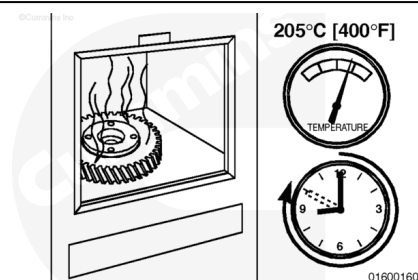
To reduce the possibility of severe burns, wear protective gloves when installing the heated ring gear.

⚠ CAUTION ⚠

Do not exceed the specified time or temperature. Damage to the ring gear and ring gear teeth can result.

⚠ CAUTION ⚠

Do not attempt to install the ring gear without heating it. The ring gear will be damaged or broken.



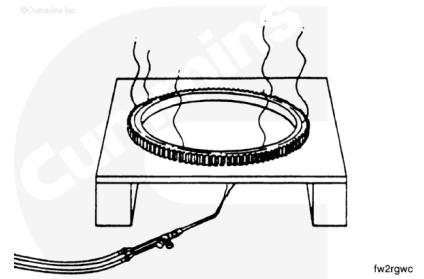
Preheat the oven to an adjusted temperature of 205°C [400°F].

Heat the ring gear in the oven for a **minimum** of 40 minutes but **not** more than a maximum of one hour.

The inner diameter of the ring gear will become larger and simplify the installation of the ring gear on the flywheel.

⚠ WARNING ⚠

To reduce the possibility of severe burns, wear protective gloves when installing the heated ring gear.



Use a heating flame to heat the gear if an oven is not available.

Use a temperature indicator marker (Templistik®), Part Number 3165163, or equivalent, to check the temperature of the gear.

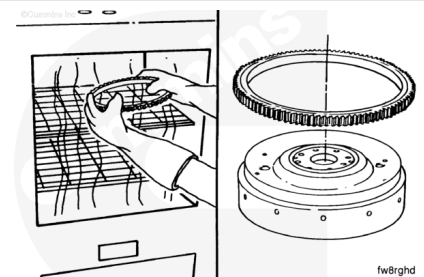
Heat the gear to 232°C [450°F].

A more even temperature is obtained by placing the ring gear on a metal plate and then heating the bottom side of the plate with the torch. Do **not** exceed the specified temperature.

⚠ WARNING ⚠

To reduce the possibility of severe burns, wear protective gloves when installing the heated ring gear.

Remove the ring gear from the oven.

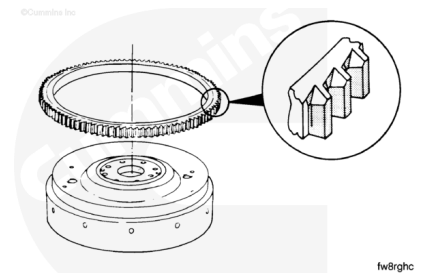


⚠ WARNING ⚠

To reduce the possibility of severe burns, wear protective gloves when installing the heated ring gear.

⚠ CAUTION ⚠

Allow the ring gear to air cool. Do not use water or oil to reduce the cooling time. Damage to the ring gear can result.



The part number is on the same side as the bevel.

Position the ring gear so that the bevel is positioned toward the crankshaft edge of the flywheel as shown.

Install the ring gear.

When installed correctly, the bevel machined at the root of each gear tooth will be toward the starting motor when the flywheel is installed.

